

1. SSH with User + Root Password Authentication

Question

You are assigned to configure an SSH server on a Linux machine.

Perform the following:

- Install and enable the SSH service
- Allow **password authentication**
- Allow **root login using password**
- Create a user admin1 with password Admin@123
- Verify SSH login for:
 - normal user (admin1)
 - root user

```
# =====
# SSH SERVER CONFIGURATION LAB
# =====
# -----
# QUESTION:
# Configure SSH server with:
# - Install & enable SSH
# - Allow password authentication
# - Allow root login
# - Create user admin1
# - Verify login (user + root)
# -----
# =====
# STEP 1: SEARCH PACKAGE
# =====
echo "Search SSH related packages"
dnf search ssh
# Used to find available SSH packages
```

```
# =====
# STEP 2: IDENTIFY PACKAGE NAME
# =====
echo "Identify package providing sshd service"
dnf provides */sshd
# Maps service (sshd) → package (openssh-server)
# =====
# STEP 3: CHECK IF INSTALLED
# =====
echo "Check if openssh-server is installed"
rpm -q openssh-server
# rpm -q = query installed package
# =====
# STEP 4: INSTALL PACKAGE
# =====
echo "Install SSH server package"
dnf install openssh-server -y
# -y = auto confirm
# =====
# STEP 5: VERIFY INSTALLED FILES
# =====
echo "List all files installed by package"
rpm -ql openssh-server
# -ql = query list all files
echo "List config files"
rpm -qc openssh-server
# -qc = query config files
# =====
# STEP 6: CHECK SERVICE STATUS
# =====
```

```
echo "Check SSH service status"
systemctl status sshd
# Shows running/inactive state
echo "Check if service enabled on boot"
systemctl is-enabled sshd
# =====
# STEP 7: START & ENABLE SERVICE
# =====
echo "Start SSH service"
systemctl start sshd
echo "Enable SSH service at boot"
systemctl enable sshd
# =====
# STEP 8: FIND SYSTEM IP
# =====
echo "Check IP address"
ip a
# Needed for SSH connection
# =====
# STEP 9: EDIT SSH CONFIGURATION
# =====
echo "Edit SSH config file"
vim /etc/ssh/sshd_config
# Make following changes inside file:
# -----
# PasswordAuthentication yes
# PermitRootLogin yes
# -----
# Save and exit (:wq)
# =====
```

```
# STEP 10: VALIDATE CONFIG
# =====

echo "Check SSH config syntax"
sshd -t

# Prevents errors before restart
# =====

# STEP 11: RESTART SERVICE
# =====

echo "Restart SSH service"
systemctl restart sshd

# =====

# STEP 12: CREATE USER
# =====

echo "Create user admin1"
useradd admin1

echo "Set password"
passwd admin1

# Enter: Admin@123
# =====

# STEP 13: VERIFY SSH LOGIN
# =====

echo "Login as normal user"
ssh admin1@<server-ip>

echo "Login as root user"
ssh root@<server-ip>

# =====

# STEP 14: EXTRA VERIFICATION
# =====

echo "Check which package owns sshd"
rpm -qf $(which sshd)
```

```
echo "View service file"
systemctl cat sshd
# =====
# NOTES / CONCEPTS
# =====
# rpm = low-level package manager
# dnf = high-level package manager
# sshd = SSH daemon (server)
# ssh = client command
# Config file location:
# /etc/ssh/sshd_config
# Service file location:
# /usr/lib/systemd/system/sshd.service
# =====
# END OF LAB
# =====
```

```
root ~
❯ rpm -qc openssh-server
/etc/pam.d/sshd
/etc/ssh/sshd_config
/etc/ssh/sshd_config.d/40-redhat-crypto-policies.conf
/etc/ssh/sshd_config.d/50-redhat.conf
/etc/sysconfig/sshd
```

```
root@localhost:~# useradd admin1
root@localhost:~# passwd admin1
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
root@localhost:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:b5:9b:60 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    altname enx000c29b59b60
    inet 192.168.220.131/24 brd 192.168.220.255 scope global dynamic noprefixroute ens160
```

```
root@localhost:~ - vim /etc/ssh/sshd_config

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes

#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
```

2. Disable Password & Use Public/Private Key Authentication

Question

Your organisation wants to improve SSH security.

Perform the following:

- Disable **password authentication**
- Configure SSH login using **public/private key**
- Create user secureuser
- Generate SSH key pair on client
- Copy public key to server
- Ensure only key-based login works

Expected Concepts

- Generate key:

ssh-keygen

- Copy key:

ssh-copy-id secureuser@<server-ip>

- SSH config changes:

PasswordAuthentication no

PubkeyAuthentication yes

- Restart service
- Verify login (should NOT ask password)

```
# =====
# SSH HARDENING LAB – KEY BASED AUTHENTICATION (ED25519)
# =====
# -----
# QUESTION:
# Disable password authentication and configure SSH using
# public/private key authentication for user secureuser
# -----
# =====
# STEP 1: CREATE USER (SERVER)
# =====
echo "Create user secureuser"
useradd secureuser
echo "Set password (temporary)"
passwd secureuser
# =====
# STEP 2: GENERATE SSH KEY (CLIENT)
# =====
echo "Generate key pair (ED25519)"
ssh-keygen -t ed25519
# Files created:
# ~/.ssh/id_ed25519 → Private key
# ~/.ssh/id_ed25519.pub → Public key
echo "Verify keys"
ls ~/.ssh/
# =====
# STEP 3: COPY PUBLIC KEY TO SERVER
# =====
echo "Copy public key to server"
# Recommended method:
```

```

ssh-copy-id secureuser@192.168.220.130
# OR manual method (your approach corrected):
scp ~/.ssh/id_ed25519.pub
secureuser@192.168.220.130:/home/secureuser/
# Then on server:
mkdir -p /home/secureuser/.ssh
cat /home/secureuser/id_ed25519.pub >>
/home/secureuser/.ssh/authorized_keys
# Fix permissions (VERY IMPORTANT):
chown -R secureuser:secureuser /home/secureuser/.ssh
chmod 700 /home/secureuser/.ssh
chmod 600 /home/secureuser/.ssh/authorized_keys
# =====
# STEP 4: VERIFY KEY LOGIN BEFORE HARDENING
# =====
echo "Test login"
ssh secureuser@192.168.220.130
# Expected:
# ✓ No password required
# =====
# STEP 5: EDIT SSH CONFIG (SERVER)
# =====
echo "Edit SSH configuration"
vim /etc/ssh/sshd_config
# Make sure these lines are set:
# -----
PasswordAuthentication no
PubkeyAuthentication yes
PermitRootLogin yes
# -----
# Save and exit (:wq)

```

```
# =====
# STEP 6: VALIDATE CONFIG
# =====

echo "Check config syntax"
sshd -t

# =====
# STEP 7: RESTART SSH SERVICE
# =====

echo "Restart SSH service"
systemctl restart sshd

# =====
# STEP 8: FINAL VERIFICATION
# =====

echo "Login using key (no password expected)"
ssh secureuser@192.168.220.130

# =====
# STEP 9: VERIFY CONFIG FILE LOCATION
# =====

echo "Check SSH config file"
rpm -qc openssh-server

# =====
# IMPORTANT FILES
# =====
# ~/.ssh/id_ed25519 → Private key (client)
# ~/.ssh/id_ed25519.pub → Public key
# /home/user/.ssh/authorized_keys → Server-side auth
# =====
# SECURITY NOTES
# =====
```

```
# Always test SSH key login BEFORE disabling password
# Wrong permissions will cause "Permission denied"
# authorized_keys must be 600
# .ssh directory must be 700
```

```
# Authentication:

#LoginGraceTime 2m
#PermitRootLogin yes

#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/author
ized_keys2
# but this is overridden so installations will only check .ssh/auth
orized_keys
AuthorizedKeysFile          .ssh/authorized_keys

#AuthorizedPrincipalsFile none
```

-----X-----

NOT FOR EXAM

3. SSH Login Using .pem Key (Cloud Scenario)

Question

You are given a .pem file to access a remote cloud server.

Perform the following:

- Set correct permissions on .pem file
- Connect to remote server using .pem key
- Login as user ec2-user (or ubuntu)
- Troubleshoot permission issues if login fails